Problem 1, first time:

Baker's total dough = 12.5 kg

Dough for bread = $3\frac{1}{4}$ kg = $\frac{(4\times3)+1}{4}$ kg = $\frac{13}{4}$ kg = 3.25 kg

Dough for cookies = 4.85 kg

Remaining dough = 12.5 kg − (4.85 kg + 3.25 kg) = 12.5 kg − 8.1 kg = 4.4 kg

$$
\begin{array}{r}
3.25 \\
4\,\overline{)\,13} \\
\underline{12} \\
010 \\
\underline{08} \\
020 \\
\underline{20} \\
00
\end{array}
$$

÷

$$
\begin{array}{r}
4.85 \\
+\ 3.25 \\
\hline
8.10
\end{array}
= 8.1
$$

$$
\begin{array}{r}
12.5 \\
-\ 8.1 \\
\hline
04.4
\end{array}
= 4.4
$$

Problem 2, first time:

Original_road_lenth = 10 miles

Construction_team_completed_old = $\frac{2}{5}$ x 10 miles

= 0.4 x 10 miles

= 4 miles

this is not programming!

write meaningful phrases

No ' ' s !

Construction_team_completed_new = $\frac{1}{4}$ x 10 miles

= 0.25 x 10 miles

= 2.5 miles

Remaining_road = 10 miles − (4 miles + 2.5 miles)

= 10 miles − 6.5 miles

= 4.5 miles

3.5

$5\overline{)2}$ = 0.4

$4\overline{)1}$ = 0.25

4 + 2.5 = 6.5

10 − 6.5 = 4.5

Problem 3, First time:

A_bank_account_balance = \$123.45

A_machine_deposits amount = \$100

Machine_withdraws amount = \$10^{-1}

= \$ $\frac{1}{10}$

= ¢10

$$\begin{array}{r} 123.45 \\ +\ 1.00 \\ \hline 124.45 \end{array}$$

$$\begin{array}{r} 12445 \\ -\ 10 \\ \hline 12435 \end{array}$$

New_balance

= (\$123.45 + \$100) - ¢10 ↓ = 1\$

= \$124.45 - ¢10

= ¢12445 - ¢10

= ¢12435

= \$124.35 ✓

Problem 4:

Length of fabric = 8.75 m

No. of pieces = 5

Length of each piece.

$= 8.75 \text{ m} \div 5$

$= 1.75$

Simple fraction $= \frac{175}{100}$ (GCD = 25)

$= \frac{25 \times 7}{25 \times 4}$

$= \frac{7}{4}$ ✓

$$\begin{array}{r} 2 \\ 25 \\ \times \quad 4 \\ \hline 100 \end{array} \div$$

$$\begin{array}{r} 1.75 \\ 5\overline{)8.75} \\ 5 \\ \hline 37 \\ 35 \\ \hline 025 \\ 25 \\ \hline 000 \end{array}$$

GCD (175, 100):

$$\begin{array}{r} 175 \\ 100\overline{)175} \\ 100 \\ \hline 075\overline{)100} \\ 75 \\ \hline 025\overline{)75} \\ 75 \\ \hline 00 \end{array}$$

$$\begin{array}{r} 3 \\ 25 \\ \times \quad 7 \\ \hline 175 \end{array}$$

Problem 5:

Mary's flour $= 5\frac{1}{3}$ cups

$= \frac{5 \times 3 + 1}{3}$ cups

$= \frac{16}{3}$ cups

Recipe flour $= 2\frac{2}{3}$ cups

flour for recipe $= \frac{2 \times 3 + 2}{3}$ cups

$= \frac{8}{3}$ cups

Recipe twice $= \frac{8}{3}$ cups $+ \frac{8}{3}$ cups

flour for making recipe twice $= \frac{16}{3}$ cups

Remaining $= \frac{16}{3}$ cups $- \frac{16}{3}$ cups

$= 0$ cups

Problem 6:

People attended on Friday $= 4^3$
$= 4 \times 4 \times 4$
$= 64$

People attended on Saturday $= 5^2$
$= 5 \times 5$
$= 25$

total people $= 64 + 25$
$= 89$

$$\begin{array}{r} 64 \\ \times\ 25 \\ \hline 89 \end{array}$$

Problem 7:

CMXLIX = (1000 − 100) + (100 − 50) + (10 − 1)

= 900 + 40 + 9

= 949

Days in 3×10^2 years

= 3 × 100 × 365

= 1095 × 100

= 109500

because 1 year = 365 days

Answer = CMXLIX < Days in 3×10^2 years

$$\begin{array}{r} 365 \\ \times\ 3 \\ \hline 1095 \end{array}$$

Problem 8:

$\sqrt[3]{216} = 6$

$\sqrt{64} = 8$

$8 \times 6 = 48$

$12 - 5 = 7$

$48 - 7 = 41$

$6 \times 6 \times 6 = 216$

$$\begin{array}{r} 33 \\ \times\ 66 \\ \hline 216 \end{array}$$

$8 \times 8 = 64$

Problem 9.

$$2^{-2} = \frac{1}{2^2}$$

$$= \frac{1}{4}$$

$$\left(\frac{1}{3}\right)^2 = \frac{1}{3} \times \frac{1}{3}$$

$$= \frac{1}{9}$$

$$\frac{1}{4} + \frac{1}{9} = \frac{1}{4} \times 1 + \frac{1}{9} \times 1$$

$$= \frac{1}{4} \times \frac{9}{9} + \frac{1}{9} \times \frac{4}{4}$$

$$= \frac{9 + 4}{36}$$

$$= 13 \div 36 \quad \left(\frac{13}{36}\right)$$

$$\frac{1}{3} \times \frac{1}{3} = \frac{1 \times 1}{3 \times 3} = \frac{1}{9}$$

Problem 10:

LXXXIII = 50 + 10 + 10 + 10 + 1 + 1 + 1

= 83

83 − 18 = 65

Roman numeral = LXV

Problem 11:

GCD = 72

144 = 72 × 2

216 = 72 × 3

Simplified fraction $= \frac{2}{3}$

$$= \frac{2 \times 72}{3 \times 72} = \frac{2}{3}$$

$$\begin{array}{r} 144 \overline{)216} \; 1 \\ \underline{144} \\ 72 \overline{)144} \; 2 \\ \underline{144} \\ 0 \end{array}$$

Problem 12:

$$\frac{7}{18} + \frac{5}{12} = \frac{7}{18} \times 1 + \frac{5}{12} \times 1$$

$$= \frac{7}{18} \times \frac{2}{2} + \frac{5}{12} \times \frac{3}{3}$$

$$= \frac{14+15}{36}$$

$$= \frac{29}{36}$$

$$\frac{29}{36}$$

What's the point of these boxes?

LCM:

18 : 18, 36, 54, 72, 90, 108 ...

12 : 12, 24, 36, 48, 60, 72, ...

LCM = 36

$18 = 2 \times 3 \times 3$

$12 = 2 \times 2 \times 3$

$$LCM = \frac{18 \times 12}{GCD(18,12)}$$

$LCM = 2^2 \times 3^2$

$= 36$

problem 13:

Given number = 1, 2 3 4, 5 x 0

Missing digit = x

Number % 9 = 0

Sum of digits % 9 = 0

Sum of digits = 1 + 2 + 3 + 4 + 5 + x + 0

= (15 + x) % 9 = 0

x = 3

Problem 14:

LD: Last Digit

$$LD(7^{45} - 3^{22}) = LD(7^{45}) - LD(3^{22})$$

$$LD(7^{45}) = LD(7^{45 \% 4}) \qquad LD(3^{22}) = LD(3^{22 \% 4})$$

$$= LD(7^{1}) \qquad = LD(3^{2})$$

$$= 7 \qquad = 9$$

$$7 - 9 \sim 17 - 9 \qquad (7^{45} > 3^{22})$$

$$= 8$$

Problem 15:

PF: Prime Factorization

$PF(72) = 8 \times 9$

$= 2 \times 2 \times 2 \times 3 \times 3$

$= 2^3 \times 3^2$

$PF(108) = 2 \times 2 \times 3 \times 3 \times 3$

$= 2^2 \times 3^3$

Distinct Prime Factors

$= 2, 3$

2	1 0 8
2	5 4
3	2 7
3	9
	3

List of all factors:

See AI notes

$(2^0, 2^1, 2^2) \quad (3^0, 3^1, 3^2)$

3×3

$= 9$ Common factors

Problem 16:

Bus A return time (A) = 45 min

Bus B return time (B) = 60 min

Same return time = LCM(A, B)

LCM(A, B) = LCM(45, 60)

$$= \frac{45 \times 60}{GCD(A, B)}$$

$$= \frac{45 \times 60}{15}$$

$$= \frac{45}{15} \times \frac{60}{1}$$

$$= 3 \times 60$$

$$= 180 \text{ min}$$

GCD =

$$\begin{array}{r|l} 45 & 60 \\ & 45 \\ \hline 15 & 45 \\ & 45 \\ \hline & 00 \end{array} \quad 1 \qquad 4 \qquad 3$$

Problem 17: I need help with Geometry.

Total Area = length × width

$= 4\frac{1}{2}\,m \times 2.5\,m$

$= \frac{9}{2} \times \frac{5}{2}\ m^2$

$= \frac{45}{4}\ m^2$

Tomato area $= \frac{1}{3} \times$ (total area)

$= \frac{1}{3} \times \frac{45}{4} = \frac{1}{\cancel{3}} \times \frac{\cancel{3} \times 15}{4}$

$= \frac{15}{4}\ m^2$

$= 3.75\ m^2$

problem 18

$GCD = 2^2 \times 3^2 \times 5^0 \times 7^0$

$= 2 \times 2 \times 3 \times 3$

$= 36$

$LCM = 2^3 \times 3^3 \times 5^1 \times 7^1$

$= 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 7$

$= 10 \times 4 \times 9 \times 21$

$= 360 \times 21$

$= 7560$

$$
\begin{array}{r}
\overset{1}{3}\,6\,0 \\
\times \quad 2\,1 \\
\hline
3\,6\,0 \\
7\,2\,0\,0 \\
\hline
7\,5\,6\,0
\end{array}
$$

problem 19:

left side $= 4.5 \times (10.249.8)$

$= 4.5 \times 20$

$= 4.5 \times 10 \times 2$

$= 45 \times 2$

$= 90$

right $= (4.5 \times 10.2) + (4.5 \times 9.8)$

$= 45.90 + 44.10$

$= 90$

$$\begin{array}{r} 15 \times 98 \\ \hline 100 \end{array} \qquad \begin{array}{r} 10.2 \\ 9.8 \\ \hline 20.0 \end{array}$$

$= 1500 - 30$

$= 1470$

$$\begin{array}{r} 1470 \\ \hline 100 \end{array}$$

$= 4410$

$$\begin{array}{r} 4410 \\ \hline 100 \end{array}$$

$$\begin{array}{r} 147 \\ \times \quad 3 \\ \hline 441 \end{array}$$

$$\begin{array}{r} 45 \\ \times 10 \\ \hline 450 \end{array} \times \begin{array}{r} 102 \\ 10 \\ \hline 1020 \end{array} = \begin{array}{r} 4590 \\ 1000 \end{array}$$

problem 20:

$$\frac{1}{5} + 0.1 \times 10^1 = \frac{1}{5} + 1$$

$$= 0.2 + 1$$

$$= 1.2$$

Roman numerals (1.2) = I

(Roman numerals are for integrs. Rounded 1.2 to 1.)